

Deployment of an AI-Assisted Hate Speech Early-Warning & Legal Triage System for Gilgit-Baltistan

Year 1 Program - PKR 29,894,666

Submitted to	The Home Secretary Home & Prison Department Government of Gilgit-Baltistan, Gilgit
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1. Executive Summary

Gilgit-Baltistan faces a documented risk of sectarian and ethnic tension being amplified through social media. Monitoring this content manually is not feasible at the volume and speed at which it now circulates, and it is multilingual - spanning Urdu, Roman Urdu, English, and regional languages including Shina, Balti, and Burushaski.

This proposal offers the Home & Prison Department a decision-support system that identifies potentially unlawful online content, groups coordinated campaigns, maps flagged material to the specific provisions of Pakistani law that may apply, and presents each case to a human officer for decision. The system does not, and structurally cannot, take enforcement action on its own.

A working prototype already exists. The classification model, the multi-agent processing pipeline, the case-file store, and the officer review dashboard are built and functioning. This proposal funds the work required to bring that prototype to a standard fit for departmental use: validation on Gilgit-Baltistan's own languages, security hardening, legal verification of the statutory mapping, on-premises infrastructure, and training of departmental officers.

The program has been costed to fit within the expenditure ceiling previously approved by the Home Secretary. The total for Year 1 - covering all development, hardware, and twelve months of subscription charges - is PKR 29,894,666.

Summary of Costs

Item	Cost (PKR)
Stage 1 — Validation pilot only (3 months, 2 districts)	6,931,375
Year 1 Complete Program - development, hardware & 12 months' subscriptions*	29,894,666
Indicative annual O&M, Year 2 onward (separate future approval)	10,626,264

** The Year 1 Complete Program figure is inclusive. It contains the Stage 1 pilot as its first three months, all subsequent development, the hardware purchase, and twelve months of subscription charges. The pilot is not an additional cost on top of it.*

Recurring charges after Year 1 are shown for planning transparency only. In line with the Department's direction, they are not sought under this approval and will be the subject of a separate submission closer to the time, when lower-cost technical options may be available.

2. Background & Problem Statement

The Department's current capacity to detect inflammatory online content depends on manual monitoring and public complaints. This produces three operational gaps:

- **Latency.** Content that incites sectarian hostility can circulate widely before it is noticed. Early identification materially changes the options available to the administration.
- **Language coverage.** A significant share of local online discourse is in Roman Urdu or regional languages, which general-purpose commercial moderation tools handle poorly or not at all.
- **Evidentiary consistency.** When a case is escalated, the legal basis is assembled manually and inconsistently. Officers require the applicable statutory provision identified alongside the content itself.

A fourth consideration governs the whole design. In a region with genuine sectarian sensitivity, a system that produces false accusations is worse than no system at all. Any deployment must therefore be built so that machine output is advisory and a human officer remains accountable for every decision. That principle is architectural here, not a policy statement added afterwards.

3. Proposed Solution - System Architecture

The system is a pipeline of specialized components, each performing one task and passing an enriched case record to the next. Processing terminates, in every case, in a human review queue.

3.1 Ingestion Agent

Collects publicly available posts through a licensed third-party data provider (Apify). No direct or unauthorized scraping is performed. The interface is pluggable so additional sources - news feeds, tip lines, other platforms - can be added without altering downstream logic.

3.2 Classification Agent

Wraps a fine-tuned XLM-RoBERTa transformer model that classifies text as neutral, offensive or hate speech across English, Urdu, and Roman Urdu, returning a confidence score and per-class probabilities.

3.3 Sarcasm Adjustment Agent

Applies only where classifier confidence falls in an ambiguous band (0.40–0.70), providing a secondary signal to reduce misclassification of ironic or rhetorical statements.

3.4 Clustering Agent - Coordinated Campaign Detection

Groups semantically similar flagged posts using multilingual sentence embeddings. Where three or more near-identical posts appear across distinct accounts, the cluster is flagged as a possible coordinated campaign — an operationally distinct signal from isolated individual speech.

3.5 Legal Mapping Agent

Matches flagged content against a curated reference table of statutory provisions and attaches the applicable citation to the case file. Provisions currently mapped include:

- Pakistan Penal Code, Section 153-A - promoting enmity between groups
- Pakistan Penal Code, Section 295-A - deliberate acts to outrage religious feelings
- Prevention of Electronic Crimes Act 2016 - offences committed through electronic communication
- Constitution of Pakistan, Article 20 - freedom of religion, recorded as the competing right to be weighed, not solely the offence to be prosecuted

Where no verified provision applies - including Gilgit-Baltistan-specific local statutes not yet mapped - the system states this explicitly rather than proposing a citation. This is a deliberate safeguard against fabricated legal authority.

3.6 Escalation Agent & Human Review Queue

Assembles the complete case file and places it in the officer review queue. Every case file is written with a mandatory human-review flag enforced at the database level by a constraint that rejects any record

attempting to bypass it. There is no code path, and no configuration setting, by which the system can act autonomously.

3.7 Officer Dashboard

A web application providing the review queue, district and platform analytics, a manual text-check facility, the legal reference framework, and a published data-handling policy. Officers confirm, dismiss or escalate each case, and every decision is recorded with the officer's identity, timestamp and reasoning.

4. Scope Calibrated to the Approved Envelope

This proposal has been prepared against the expenditure ceiling previously approved by the Home Secretary. Delivering within that ceiling required genuine reductions in scope, not a discount on the same scope. Setting those reductions out plainly is essential: an approval granted on an unstated assumption of wider scope would fail during delivery.

The table below states what the Year 1 budget delivers and what has been deferred.

Capability	Year 1 (this proposal)	Deferred
Core agent pipeline & officer dashboard	Included in full	—
Human review, audit trail & accountability constraint	Included in full	—
Legal mapping with advocate validation	Included in full	—
Server infrastructure	Single server	High-availability pair
Disaster recovery site	Not included	Deferred — off-site backup only
Ingestion capacity	~100,000 posts/month	300,000–500,000 (Business tier)
GB annotation corpus	~25,000 samples	100,000 samples
Sarcasm detection	Basic trained model	Fully validated model
Security testing	One penetration test	Second round, ISO 27001 readiness
Independent bias audit	Not included	Deferred
Review officer workstations	4	15
Officer training	3 regional sessions	Per-district programme
Support cover	Business hours, L1	Extended hours L1 / L2

4.1 Consequences the Department should note

- No automatic failover. The system runs on a single server. A hardware fault causes an outage lasting hours to days depending on spares availability, rather than seconds. This is the most significant consequence of the reduced envelope and is accepted knowingly.
- A volume ceiling of approximately 100,000 posts per month. Beyond this, ingestion costs rise, and a subscription-tier upgrade becomes necessary. Volume is monitored and reported so the Department is never surprised by an overage.
- A smaller training corpus means smaller accuracy gains. Roughly 25,000 labelled GB samples will improve performance on local dialects, but by less than the 100,000-sample program originally scoped.

- Deferred assurance items. Independent bias audit, a second penetration test round and ISO 27001 readiness are outside this budget. They are recommended before the system informs any prosecution decision.

None of these reductions touch the safeguards. Human review, the audit trail, the accountability constraint and the legal-mapping validation are funded in full, because they are the elements that make the system defensible.

5. Governance, Legal & Human Rights Safeguards

A system of this nature operates close to constitutionally protected expression. The following safeguards are built into the design and are proposed as binding conditions of deployment:

- Human decision mandatory. No content is acted upon automatically. The constraint is enforced in the data layer, not merely in policy.
- Advisory legal output. Statutory mapping is an assistive drafting aid for a legally trained reviewer to confirm. The system does not issue legal determinations, and the interface states this on every case.
- No fabricated citations. Where a provision cannot be verified, the system records “not yet mapped”.
- Full audit trail. Every decision record the deciding officer, timestamp, decision, and written reasoning, supporting subsequent departmental or judicial scrutiny.
- Published transparency policy. A data-handling policy stating what is collected, why, and the limitations of the system is included in the application itself.
- Data sovereignty. On-premises deployment keeps citizen data within departmental premises; no citizen content is stored with a foreign cloud provider.
- Proportionality. Only publicly available content is processed. The system does not access private messages, accounts or devices.

6. Present Status of the System

The Department is entitled to an accurate account of what exists today and what remains to be built.

Component	Present Status	Notes
Hate-speech classifier (XLM-RoBERTa)	Operational	Trained & deployed; not yet validated on GB content
Multi-agent orchestration	Operational	Conditional routing tested end-to-end
Escalation & human review queue	Operational	Human review enforced at database level
Legal provision mapping	Functional — requires validation	Keyword matching; needs advocate sign-off
Coordinated-campaign clustering	Simplified implementation	Sentence embeddings + cosine similarity
Sarcasm detection	Placeholder (rule-based)	Not a trained model; replacement funded in Year 1
District-level geo-tagging	Not available	No reliable location source in ingested data
Authentication & role-based access	Not implemented	Funded in Year 1; required before production
Storage layer	SQLite (development)	Migration to PostgreSQL funded in Year 1

The published accuracy figure for this model architecture (95.7%) was obtained on its original academic test set. It has not been validated on Gilgit-Baltistan content, and real-world accuracy on local dialects should be expected to be lower until the retraining funded here is complete. No accuracy guarantee is offered on that basis; Stage 1 exists precisely to measure it.

7. Implementation Approach

The programme runs as a single twelve-month engagement with an early validation stage. Stage 1 establishes measured accuracy on the Department's own content within the first four months, giving an evidence-based review point before the larger share of expenditure is committed.

Stage	Period & Scope	Objective	Cost (PKR)
Stage 1	Months 1–3 - Gilgit & Skardu	Measure real accuracy on GB content; establish officer workflow	6,931,375
Stage 2	Months 4–12 - remaining districts	Model retraining, security hardening, rollout & training	22,963,291
Year 1 Complete Program	12 months	Development, hardware & 12 months' subscriptions	29,894,666

Stage 2 is shown as the balance of the Year 1 program (PKR 29,894,666 less PKR 6,931,375). The two stages together are the single approved figure, not two separate purchases.

8. Cost Plan A — Stage 1 Validation Pilot

Scope: three months, Gilgit and Skardu districts, a single GPU workstation (carried forward into the full program), initial GB sample labelling, and a measured accuracy report delivered to the Department.

Cost Head	Amount (PKR)
Human resources - 7 roles, part-time over 3 months	3,497,500
Hardware - GPU workstation, UPS, 2 review stations (carried into full program)	1,580,000
Software & subscriptions (3 months)	227,160
Data & annotation - initial GB sample set	350,000
Training - single session, 2 districts	150,000
Travel & field	250,000
Base Cost	6,054,660
Administrative overhead (6%)	363,280
Contingency (8%)	513,435
TOTAL — STAGE 1 VALIDATION PILOT	6,931,375

Stage 1 total: PKR 6,931,375 (approximately USD 24,755)

This figure is contained within the Year 1 program total and is presented separately only so the Department can see the cost of the evidence-gathering stage in isolation.

9. Cost Plan B - Year 1 Complete Program

Scope: twelve months covering all development work, the hardware purchase, twelve months of subscription and service charges, GB corpus collection and model retraining, security hardening, legal validation, officer training and rollout across the districts.

9.1 Human Resources

Role	Rate / month	FTE	Months	Amount (PKR)
Project Manager / Technical Lead	450,000	0.50	12	2,700,000
ML / NLP Engineer	600,000	0.75	12	5,400,000
Backend Engineer (Python / FastAPI)	400,000	0.75	12	3,600,000
Frontend Engineer (Next.js / React)	350,000	0.50	8	1,400,000
DevOps / MLOps Engineer	500,000	0.25	12	1,500,000
QA / Test Engineer	250,000	0.35	8	700,000
Linguists & Annotators - Urdu, Shina, Balti (2 persons)	110,000	2.00	5	1,100,000
Legal Advisor (PPC / PECA validation)	350,000	0.15	12	630,000
Security Engineer (review & hardening)	600,000	0.15	4	360,000
Training & Change Management Officer	300,000	0.20	4	240,000
Support Engineer (L1)	220,000	0.50	6	660,000
Sub-total - Human Resources				18,290,000

9.2 Hardware (Capital Expenditure)

Item	Qty	Unit (PKR)	Amount (PKR)
Inference server - GPU (16 GB class), 128 GB RAM; also hosts application & database	1	1,600,000	1,600,000
Backup storage / NAS with scheduled off-site replication	1	350,000	350,000
Online UPS (3 kVA), rack, managed switch, firewall appliance	1	550,000	550,000
Review officer workstations (dual-monitor)	4	140,000	560,000
Installation, cabling & commissioning	1	150,000	150,000
Sub-total — Hardware (CapEx)			3,210,000

9.3 Software & Subscriptions — 12 Months

Item	USD total	Amount (PKR)
Apify - Scale plan (USD 199 / month × 12)	USD 2,388	668,640
Apify - proxy & compute-unit overage (est. USD 100 / month)	USD 1,200	336,000
Monitoring & error tracking (USD 50 / month)	USD 600	168,000
Encrypted off-site backup (USD 30 / month)	USD 360	100,800
Domain & TLS certificates	—	40,000

Item	USD total	Amount (PKR)
Sub-total — Software & Subscriptions (12 months)		1,313,440

9.4 Data, Training, Legal & Field Costs

Item	Amount (PKR)
GB-specific labelled corpus - approx. 25,000 samples (tooling, incentives, adjudication)	900,000
Independent linguistic spot-validation - GB dialects	300,000
Data & Annotation sub-total	1,200,000
Officer training - 3 clustered regional sessions	500,000
Documentation (English & Urdu), SOPs, user manuals	300,000
Training & Capacity Building sub-total	800,000
Legal provision validation (advocate review)	400,000
Oversight framework & standard operating procedures	200,000
Legal & Governance sub-total	600,000
District deployment visits	450,000
Accommodation, per diem & logistics	250,000
Travel & Field sub-total	700,000

9.5 Year 1 Cost Summary

Cost Head	Amount (PKR)
A. Human Resources	18,290,000
B. Hardware (CapEx)	3,210,000
C. Software & Subscriptions (12 months)	1,313,440
D. Data & Annotation	1,200,000
E. Training & Capacity Building	800,000
F. Legal & Governance	600,000
G. Travel & Field	700,000
Base Cost	26,113,440
H. Administrative & compliance overhead (6%)	1,566,806
I. Contingency & risk reserve (8%)	2,214,420
TOTAL — YEAR 1 COMPLETE PROGRAMME	29,894,666

Year 1 total: PKR 29,894,666 (approximately USD 106,767)

This figure includes all one-time development, the hardware purchase, and twelve months of subscription charges. No further expenditure is sought for the first year.

10. Recurring Charges After Year 1

Shown for planning transparency. In line with the Department's direction, these are not sought under the present approval and will form a separate submission nearer the time, by which point lower-cost technical options for ingestion and hosting may be available.

Item	Amount (PKR / year)
Skeleton operations team - DevOps 0.2, Backend 0.25, ML 0.25, Support 0.5, PM 0.15 FTE	6,330,000
Apify subscription & overage	1,004,640
Monitoring & off-site backup	268,800
Hardware annual maintenance (8% of CapEx)	256,800
Electricity & connectivity	600,000
Annual security review	400,000
Model retraining — twice yearly	800,000
Base Recurring Cost	9,660,240
Contingency (10%)	966,024
INDICATIVE ANNUAL O&M — YEAR 2 ONWARD	10,626,264

Model accuracy degrades as language and events change. The retraining line is not discretionary maintenance - it is what prevents the system from becoming progressively less reliable while continuing to appear authoritative.

Indicative five-year total cost of ownership, assuming the above recurring profile is approved in later years: PKR 72,399,722 (approximately USD 258,570).

11. Assumptions & Exclusions

11.1 Assumptions

- Exchange rate of 1 USD = PKR 280. USD-denominated items will be re-based at the prevailing rate on the date of the purchase order.
- The Department provides a physically secure server room with reliable mains power and a dedicated internet connection at the Gilgit premises.
- The Department nominates review officers and makes them available for scheduled training.
- Necessary legal authority and departmental approval for monitoring publicly available content is obtained by the Department.
- Salary rates reflect Pakistan market benchmarks for 2026 and assume a locally based delivery team.
- Apify subscription tiers are as published at the date of this proposal and are subject to change by the vendor.
- Monitored volume remains within approximately 100,000 posts per month.

11.2 Exclusions

- Government taxes and duties (GST, withholding tax, customs duty on imported hardware) are excluded and will be charged at applicable rates.
- Civil works, server-room construction and structural electrical work.
- Recurring internet connectivity charges (assumed departmental provision).
- Any licence, subscription or access fee levied directly by a social media platform.
- Subscription and maintenance charges beyond the first twelve months.

- High-availability infrastructure, disaster-recovery site, ISO 27001 certification and independent bias audit, as set out in Section 4.
- Legal costs of any proceedings arising from a decision taken by a departmental officer.

12. Risk Register

Risk	Severity	Mitigation
False positive leading to wrongful action against a citizen	High	Mandatory human review; no autonomous path exists in the architecture; legal officer sign-off before action
Under-performance on Shina / Balti / Burushaski	High	GB corpus and retraining funded; accuracy reported per language; Stage 1 measures this before full commitment
Single-server failure causes outage	Medium-High	Accepted consequence of the budget envelope. Mitigated by next-business-day spares, nightly off-site backup and documented rebuild procedure - not by automatic failover
Ingestion volume exceeds Scale-tier capacity	Medium	Volume monitoring with alerting; tier upgrade costed and presented before any breach
Perceived as mass surveillance	High	Published transparency policy, grievance route, legal-basis disclosure
Foreign-exchange movement on USD line items	Medium	Contingency reserve; annual re-basing at purchase-order date
Key-person dependency on specialist staff	Medium	Documentation and train-the-trainer handover to departmental staff

14. Recommendation

The Department is invited to approve the Year 1 Complete Program at PKR 29,894,666, comprising all development, hardware and twelve months of subscription charges.

The program has been shaped to fit the approved ceiling honestly. Where scope had to be reduced, it has been reduced in infrastructure resilience, assurance depth and processing capacity - not in the safeguards. Human review, the audit trail, the accountability constraint and legal validation are funded in full, because they are what make the system defensible if a decision taken with its assistance is ever challenged.

The Stage 1 review point at Month 4 gives the Department measured accuracy on its own content before the greater part of the expenditure is committed, and a documented basis for continuing, adjusting or halting.

Submitted for the consideration of the Home Secretary, Home & Prison Department, Government of Gilgit-Baltistan.

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ZAIN ABBAS	Date
TECHNICAL LEAD	

ZIRCON LOGICS	
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Annexure A — Basis of Rates & Pricing Sources

All figures are indicative estimates prepared for budgetary planning. They are not a fixed-price quotation and should be confirmed at the time of procurement.

A.1 Staffing Rates

Monthly rates reflect published 2026 Pakistan market benchmarks. Artificial-intelligence and machine-learning engineering commands the highest band in the domestic market (reported range PKR 300,000–1,500,000 per month); cloud and DevOps specialisation follows (PKR 200,000–900,000). Senior software engineering roles fall in the PKR 200,000–400,000 band, with mid-level roles at PKR 100,000–250,000. Rates used here sit at or below the mid-point of the applicable band and are inclusive of employer contributions and overhead. Cost control in this proposal is achieved through reduced headcount and part-time allocation rather than below-market rates, which would not attract or retain the specialists required.

A.2 Third-Party Service Pricing

Apify subscription tiers as published in 2026: Free (USD 0), Starter (USD 29 / month), Scale (USD 199 / month), Business (USD 999 / month). Compute is billed in Compute Units, where one unit equals one gigabyte of memory for one hour, at USD 0.13–0.20 per unit depending on tier. Subscription fees function as a prepaid usage budget; consumption beyond it is billed as overage. This proposal adopts the Scale tier, which supports the stated volume ceiling. Annual pre-payment attracts a 10% discount the Department may wish to take up.

A.3 Hardware Pricing

GPU pricing is based on 2026 market rates: NVIDIA L4 at USD 2,000–3,000 and RTX A6000 at USD 3,500–4,650. This proposal specifies a 16 GB-class GPU, which is sufficient for the models deployed. It is worth stating plainly that this workload does not require datacentre-class accelerators: the classification and embedding models are comparatively small, and the specification has been sized to the actual requirement rather than over-provisioned.

A.4 Foreign Exchange

Converted at 1 USD = PKR 280. The USD/PKR rate averaged approximately 277.9 over the month to 24 August 2026, within a range of 277.4 to 281.2. A modest buffer has been applied for planning purposes.

A.5 Volume Assumptions

Year 1 assumes up to approximately 100,000 posts per month. Apify consumption and proxy costs scale with this volume and are the line items most sensitive to variation in actual usage. Volume is monitored, and any approach to the ceiling will be reported to the Department with costed options before an overage is incurred.